DIGITAL LITERACY TASK 4-A

CASE STUDY: Ransomware in Healthcare

Incident: The Change Healthcare Cyberattack (February 2024)

Source: U.S. Department of Health and Human Services (HHS) / UnitedHealth Group Testimony to Congress

What the Crime Is ?

Ransomware is a type of malware attack where attackers encrypt the victim's files, making them inaccessible. The attackers, in return, demand a "ransom" to provide the decryption key to the victim. In most cases, the attackers demand this "ransom" in cryptocurrency form.

How It Happens (Step-by-Step):

1. **Initial Access**: In the case of Change Healthcare, attackers breached the company on February 12, 2024, by entering the remote access portal by using compromised credentials. The remote access portal did not have MFA (Multi-Factor Authentication).
2. **Lateral Movement**: The attackers moved laterally in the network for nine days, identifying key systems and collecting approximately 6 terabytes of sensitive data.
3. **Deployment**: On February 21, 2024, the attackers deployed the ransomware, encrypting the systems processing 15 billion healthcare transactions annually.
4. **Extortion**: The attackers, known by the alias "BlackCat" (ALPHV), demanded a ransom. The UnitedHealth Group decided to pay a $22 million ransom in exchange for Bitcoin to restore operations and prevent leaks.

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Targeted Parties:

Healthcare providers and clearinghouses are considered prime targets, especially since they manage critical infrastructure. They are attacked because of the need for immediate compensation, which makes them more likely to pay quickly. In this particular case, the target was a "single point of failure" which processed close to 40% of all medical claims made in the United States.

Consequences :

- **Financial:** The cost of recovery and response was reported by UnitedHealth Group, which totalled more than $2.4 billion.
- **Operational:** Pharmacies could not process prescriptions, and hospitals experienced a massive cash flow crisis.
- **Human impact:** Over 100 million Americans had their private medical information stolen, with patients experiencing life-threatening delays in receiving necessary medical care.